

Abtin Olae

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EDUCATION

M.S Data Science

San Jose State University

Expected May 2027

B.S Statistics and Data Science

University of California Santa Barbara

GPA: 3.76 | Dean's Honors: Fall 2023, Spring 2024, Spring 2025

Jun 2025

PROJECTS

AI Forecast Validation | github.com/DreamingBlues/ai-models_validation

Spring 2026

- Analyzed high-dimensional atmospheric data (1M+ observations per timestep) to validate performance and of emerging AI weather models such as Nvidia FourCastNetv3
- Designed Python ETL pipelines using xarray and h5netcdf into analysis ready formats
- Performed topography and synoptic weather station analysis and created clear explanatory visualizations
- Collaborated and provided research insights to utility companies such as PG&E and SCE
- First-authored research manuscript and submitted for publication peer review at *MDPI: Forecasting*

Hyperspectral Vegetation Mapping | github.com/DreamingBlues/drone-vegetation-mapping

Spring 2026

- Processed a 430-band aerial hyperspectral image using Zarr chunking, manual land-cover masks, and balanced stratified sampling across water, soil, canopy, and grass classes
- Applied PCA, spectral correlation analysis, and wavelength-distance scoring to reduce the dataset
- Trained and compared Random Forest classifiers, achieving 92.7% accuracy with the seven-band model while reducing training time by 92.6% and preserving most of the full 430-band model's performance

SKILLS & CERTIFICATIONS

Languages: Python, R, SQL, Bash, Git, Linux/Unix

Data & ML: NumPy, Pandas, TensorFlow, Scikit-learn, PyTorch, XArray

Big Data & Cloud: Apache Spark, Databricks, Google Cloud Platform, AWS S3

Visualization: Matplotlib, Seaborn, Cartopy, Metpy, Earthkit

EXPERIENCE

Graduate Student Researcher

Sep 2025 – Present

Wildfire Interdisciplinary Research Center, San Jose State University

- Collaborated in an interdisciplinary research environment, contributing to multiple projects and quickly adapting to new tools, methods, and research domains
- Contributed to FireBench, an open-source Python benchmarking framework, by designing ETL pipelines for weather and fuel-moisture data
- Led privately funded research projects, overseeing project analysis and presenting weekly progress briefings to stakeholders

Software Engineering Intern

Jun 2023 – Aug 2023

Frugal Innovation Hub, Santa Clara University

- Designed mobile application using Figma
- Developed mobile application features using Flutter and Dart.
- Integrated Firebase cloud databases to support user tracking and analytics.

ACTIVITIES

Vice President and Co-Founder

Nov 2021 - May 2023

IEEE Student Chapter, Ohlone College

- Led 30+ students and organized technical project showcases and conference participation
- Collaborated on interdisciplinary project, mapping UAV-based gas detection and vertical sounding systems